

SUMMARY

Electrical engineer with a passion for robotics and embedded electronics. Worked in R&D for closer to 3 years with experience in micro-controllers, unmanned systems, software defined radios and EdTech (education technology).

EDUCATION

Tandon School of Engineering, New York University , Brooklyn, New York	<i>May 2020</i>
Master of Science in Electrical and Electronics Engineering	
Sri Sairam Engineering College, Anna University , Chennai, India	<i>May 2018</i>
Bachelor of Engineering in Electronics and Instrumentation Engineering	

SKILLS

- Proficient in **Python** and **C** programming
- Experience with **Fusion 360**, **KiCAD** and **MATLAB**
- Knowledge of **Raspberry Pi**, **Arduino**, **SDRs**, **Pixhawk** and **Ardupilot**
- Familiar with **SPI**, **I2C**, **UART** protocols

PROFESSIONAL EXPERIENCE

Building Momentum LLC, Virginia, USA *January 2022 – August 2024*

Senior Research and Development Associate

Unmanned Systems (UAV, USV)

- Developed an operational sUAS using **Pixhawk 4** microcontroller and **M8N GPS** System with the **Ardupilot** Platform
- Integrated the **Mission Planner** application with Ardupilot to setup **simple waypoint missions** and **RTL**(return to location) missions on the sUAS
- Implemented an unmanned mission on a dinghy to go drop a sensor packet about a mile into the ocean using **Pixracer** and **Ardupilot**

Software Defined Radios (SDR)

- Developed a functional radio prototype using **HackRF One SDR** and **Raspberry Pi** processor to demonstrate both transmission and reception of radio waves for utilization in the field
- Implemented the transmitting and receiving capabilities on both **PiSDR** and **Dragon OS** platforms using **GNU Radio & Python** programming

PCB Design and Milling

- Designed an IR Firefly **PCB** from the scratch using **KiCAD** software to help servicemen distinguish their own troop from their opponents
- Tested the PCB design by milling a single sided PCB on the **Bantam 4-axis Desktop Milling Machine** using the **Bantam Tools Software** and soldering the components onto the prototype board before sending it out for bulk manufacturing

Everel America Inc., Georgia, USA

January 2021 – December 2021

Application Engineer

- Collaborated with customer and supply engineers in the design work of various types of switches like **CAM switches**, **rocker switches**, **waterproof switches** and **push button switches** to meet **UL94 standard**
- Performed **testing & quality assurance** on the switches before the delivery of the finished product to the customer

Hotel Booking Cancellation Analysis

- Extracted, loaded and transformed (**ETL**) data in the database using **MySQL** and **Python** programming
- Applied different **Machine Learning** models like **Logistic Regression**, **Random Forest**, **Decision Trees** to the extracted data and performed **model optimization** using **Grid Search** to predict the rate of future cancellations ([GitHub - Akshtasuresh/Hotel-Booking-Cancellation](#))

ACADEMIC PROJECTS

Real Time Detection of Texts in Videos using Webcam

March 2020 – May 2020

- Implemented a text detection application in videos using the real time webcam data from the computer
- Used **OpenCV library** and a deep learning based text detector called **EAST** : An Efficient and Accurate Text Detector in **Python** for the video processing and text detection respectively ([GitHub - Akshtasuresh/Real-Time-Text-Detection-in-Videos-using-WebCam](#))

Swarm Robots

November 2019-December 2019

- Implemented a simulation in which a swarm of robots moves towards a goal by avoiding the obstacles in specified formations and also move objects to certain specified locations using the swarm robots by caging the object
- Used **PyBullet** as the simulation environment ([GitHub - Akshtasuresh/Swarm-Robotics](#))

Embedded Rendezvous Robot

September 2019-December 2019

- Designed a robot car that would move towards a wifi beacon avoiding the obstacles and stop at the beacon
- Figured out and coded an algorithm for deciding the path that the robot car should take based on the RSSI measurements and also for obstacle avoidance in **Arduino** for the HUZZAH ESP8266 microcontroller ([GitHub - Akshtasuresh/Embedded-Systems](#))

SLAM for Mobile Robots

February 2019-May 2019

- Worked on a simulation project wherein **SLAM** was implemented on a mobile robot, which created a map of the environment and reached its destination
- Used **VREP** as the robotics simulation environment, **MATLAB** for coding purposes and linked the MATLAB code with the VREP environment

ROBOBUD

October 2018-December 2018

- Built a robot that carries a speaker close to a person's hearing range using **Arduino UNO** and **XBEE** Modules
- Equipped the robot with obstacle avoidance using ultrasonic sensors

VOTOMETRICS

December 2017-March 2018

- Built a miniature model of a voting machine integrated with a biometric sensor using **Arduino UNO** and the Unique Identity Number of a person in order to make the voting system more efficient
- Developed a database for the same using **MYSQL** and used **HTML** for the web design

ALL IN ONE BOT

December 2015

- Designed, assembled and coded in **Embedded C** for a line follower robot along with obstacle avoidance and speed regulation using **AtMega16** controller for a robotics competition ([GitHub - Akshtasuresh/All-in-one-bot-line-follower-obstacle-avoidance-speed-regulation-: local to remote checkin](#))